Mains input (HV)

L/N 100–240 VAC –> T2A slow-blow fuse –> 10D561K MOV –> 5D-9 NTC –> HLK-20M12. Fuse and MOV as Hi-Link recommends for the module; the NTC limits inrush. 6.5 mm clearance to all SELV copper.

AC/DC 12 V 1.6 A

HLK-20M12: 85–264 VAC in, 12 V 1.6 A (20 W) out, 2.5 kV isolation. SS54 stops an external 12 V source from back-feeding the module. 470 uF polymer on +12V for load steps.

12 V terminal (in or out), reverse-polarity protected

10.5–16 V in from a 12 V battery system, or +12V out. SMBJ16CA clamps surges. AO4805 back-to-back P-FETs conduct only with correct polarity (2N7002 gate drive from a 1:2 divider); 100k from +12V lets them close when the board supplies 12 V out. Reversed: both FETs off.

12 V –> 5 V buck (LMR51430XF, 3 A, 500 kHz FPWM)

TI typical application: 2 x 4.7 uF 50 V + 100n at VIN, 100n boot, 6.8 uH (Isat 13 A, above the 6.7 A peak current limit), 2 x 22 uF out. EN tied to VIN: always on. Short CIN–VIN–GND loop and SW node over a solid GND pour.

Buck feedback divider

Vout = 0.6 V x (1 + 73.2k / 10k) = 4.99 V; 0.1 % resistors keep +5V within 1.7 %. Placed at the FB pin, sensed at the output capacitors.

5 V rail for the ESP32 board and modules (ESP_5V): buck or USB

LM66100 ideal diode from the buck (+5V) to ESP_5V, which feeds the dev board, the four HW-519s, the CAN module, the TXS0108E 5 V side and the microSD module. On USB alone the dev board feeds ESP_5V, so everything except the SIM800L and the 12 V relay output runs; the LM66100 stops USB from back-feeding the buck. Up to ~0.95 A on 12 V (1.5 A rating).

Power indicator + test points

Green LED on +5V (~3 mA)

Power flags

Mechanical (M4 holes, DIN-rail clips)

Three plated M4 corner holes (6.4 mm ring, not connected; M4 screws with a 7 mm head, ISO 4762 / ISO 14580); the fourth corner holds the mains input. Two Bud DNC-4796 steel clips for 35 mm DIN rail, M4 x 8 screws from the top; no bottom copper or THT pins under the clips.

RS-485 1 (GPIO6 RX / GPIO7 TX)

HW-519 module on 5 V; Modbus client port. Bus wires on the module terminal.

RS-485 4 (GPIO47 RX via divider / GPIO13 TX)

HW-519 module on 5 V; meter bus. RXD (5 V push-pull) through a 2.2k / 3.9k divider: 3.0–3.2 V over the 5 V range; ~40 ns edges against an 8.7 us bit at 115200 baud.

Display (1.3" SH1106 OLED, SPI)

3.3 V; shares SCK/MOSI with the CAN bus (3.3 V side). RES on GPIO4, not a strap pin.

CAN 2.0 (MCP2515 + TJA1050 module)

SPI through the level shifter: SCK GPIO8, MOSI GPIO41, MISO GPIO42, CS GPIO2, INT GPIO17. CANH/CANL and the 120R termination jumper are on the module.

ESP32-S3 dev board (GoouuuTech S3-N16R8)

Socketed (2 x 21, rows 25.4 mm) so the dev board can be replaced; its USB-C sits at the board edge for flashing and power. Left free: GPIO35–37 (octal PSRAM), GPIO0/45 (straps), GPIO19/20 (USB). GPIO46 (strap) carries LoRa DIO0, which is low after SX1278 reset. GPIO43/44 (UART0) carry SD_CS and ETH_INT.

Level shifter 3.3 V <–> 5 V (TXS0108E module)

A side 3.3 V (ESP32), B side 5 V (modules). OE pulled up: always enabled. RS-485 TXD goes direct: the HW-519 input threshold (~2 V) accepts 3.3 V logic.

RS-485 2 (GPIO15 RX / GPIO16 TX)

HW-519 module on 5 V; meter bus (DLMS, DL/T 645, Modbus).

RS-485 3 (GPIO18 RX via TXS A8/B8 / GPIO5 TX)

HW-519 module on 5 V; meter bus. RXD shifted to 3.3 V by the TXS0108E.

LoRa: Ai-Thinker Ra-01 (SX1278, 433 MHz) on a 2.54 mm adapter

2 x 1x8 sockets for the Ra-01 adapter. NSS GPIO14 (10k pull-up), DIO0 GPIO46, RESET from NET_RST (GPIO21, 10k pull-up). Antenna end at the board edge. 10u + 100n for the ~120 mA transmit current.

4G alternative: A7670 Cat-1 board (SIMCom A7670E)

Second 1x7 socket in the same position as the SIM800L, in the A7670 board pin order: 1 GND, 2 URX <– GPIO10, 3 UTX –> GPIO11, 4 PWR-K through solder jumper JP1, 5 VCC (SIM_5V; the board takes 5–12 V), 6 GND, 7 SLEEP (open). Fit one board or the other: they share the UART, the supply and the reset line. 4G keeps working where 2G networks are switched off.

SIM800L EVB V2.0 5V (2G GSM/GPRS)

1x7 socket on the board's 24 mm edge: 1 VCC, 2 GND, 3 VDD (logic reference, 3V3), 4 SIM_TXD –> GPIO11, 5 SIM_RXD <– GPIO10, 6 GND, 7 RST. The board lies flat over the socket, header pins down.

A7670 PWR-K select (solder jumper)

JP1 sets how the A7670 board is switched on. 1–2 (default, copper bridge): from NET_RST through the BAT54, so firmware presses the power key by pulsing NET_RST low. 2–3 (cut 1–2, solder 2–3): PWR-K held at GND, so the modem starts by itself at power-up. All open: for boards that switch themselves on.

Ethernet (W5500 module)

2x5 2.54 mm socket for the common W5500 module: 1 NC, 2 INT, 3 RST, 4 GND, 5 5V (not connected: 3.3 V supply only), 6 SCLK, 7 SCS, 8 MOSI, 9 MISO, 10 3.3V. CS GPIO39 (10k pull-up), RST from NET_RST (GPIO21). INT on GPIO44 (UART0 RX) through 1k, which limits the current against the dev board USB-UART bridge. Own MISO (GPIO40), separate from the CAN MISO. 22u + 100n for ~130 mA.

microSD module (SPI, 5 V)

Plug-in module with its own level shifter: 1x6 GND VCC MISO MOSI SCK CS. Its 74LVC125 never releases MISO, so MISO reaches the shared bus through a 74LVC1G125 enabled only while SD_CS is low. CS GPIO43 (UART0 TX, 10k pull-up): boot-log activity on it is harmless with SCK idle.

Input connectors (JST XH 2.50 mm, 2-pin each)

Pin 1 input, pin 2 GND: a reed or limit switch, or a push button, between the two pins.

I2C (GPIO12 SDA, GPIO48 SCL)

Header for external sensors or expanders: 1 +3V3, 2 GND, 3 SDA, 4 SCL. 4.7k pull-ups to 3V3. GPIO48 also drives the RGB LED on many ESP32-S3 dev boards.

SIM800L reset from NET_RST

BAT54WS, cathode to NET_RST: NET_RST low resets the modem together with the W5500 and Ra-01; high leaves RST to the module's own 2.8 V pull-up, so 3.3 V never reaches the pin.

Contact inputs DOOR / IN1 / IN2 (3-bit resistor DAC on GPIO1)

Three dry-contact inputs on one ADC pin as a 3-bit resistor DAC: DOOR 2.4k, IN1 5.6k, IN2 13k to IN_ADC; IN_ADC 10k up, 8.2k down, 100n. All 8 combinations stay >= 0.17 V apart with 1 % parts and 3V3 +/-2 %. Per input: 220R + 100n at the connector against ESD and noise; 2.2k pull-up gives ~1.4 mA wetting current.

SIM800L supply (2 A bursts)

From +5V through 0.1 R with 2 x 1000 uF polymer at the header: the capacitors supply the 2 A, 577 us transmit bursts, so the buck stays below its current limit and +5V within 4.87–5.13 V.

Relay output, 12 V (external relay / contactor coil, <= 300 mA)

Low-side switch for an external relay or contactor coil. GPIO38 high = on. AO3400A (RDS(on) < 48 mR at 2.5 V gate), 100R gate resistor, 100k keeps it off during reset. PTC 0.5 A / 30 V protects against a shorted coil; SS14 clamps the coil flyback.

Relay status LED

Red LED, ~2 mA while the output is on

Notes

Power
- Mains: at least 6.5 mm creepage and clearance from AC_L, AC_L_F, AC_L_NTC and AC_N to all SELV copper and to the M3 and DIN-clip metal.
- 12 V comes from the HLK-20M12 or the 12V IN/OUT terminal; the buck makes +5V for the SIM800L and, through the LM66100, ESP_5V for everything else on 5 V. On USB alone all of it runs except the SIM800L and the relay output.

Controller
- GPIO1 = IN_ADC (DOOR / IN1 / IN2), GPIO12 / GPIO48 = I2C, GPIO38 = relay output.
- Five serial devices (SIM800L, RS-485 1–4) share the three ESP32-S3 UARTs through the GPIO matrix.

Interfaces
- RS-485 ports are numbered by GPIO (RX / TX): 1 = 6 / 7, 2 = 15 / 16, 3 = 18 / 5, 4 = 47 / 13.
- Surge and ESD protection is on the HW-519 modules, at their terminals.

Network
- 3.3 V SPI bus (OLED, W5500, microSD, LoRa): SCK GPIO8, MOSI GPIO41, MISO GPIO40.

GSM
- Fit either the SIM800L EVB (2G GSM/GPRS) or the A7670 board (4G LTE Cat-1), never both.

Relay output
- The relay output runs from +12V, so it is not available on USB power alone.

Carrier for the EMS dev kit modules: ESP32-S3 board, 4x HW-519, MCP2515,
TXS0108E, OLED, W5500, microSD, Ra-01, SIM800L or A7670; 12 V: HLK-20M12
+ 12V IN/OUT terminal, buck to 5 V; I/O: DOOR/IN1/IN2, I2C, relay 12 V
DANGER: mains voltage in the bottom-right corner; 2 layers, 1.6 mm

Sheet: /
File: ems-carrier.kicad_sch

Title: EMS carrier

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